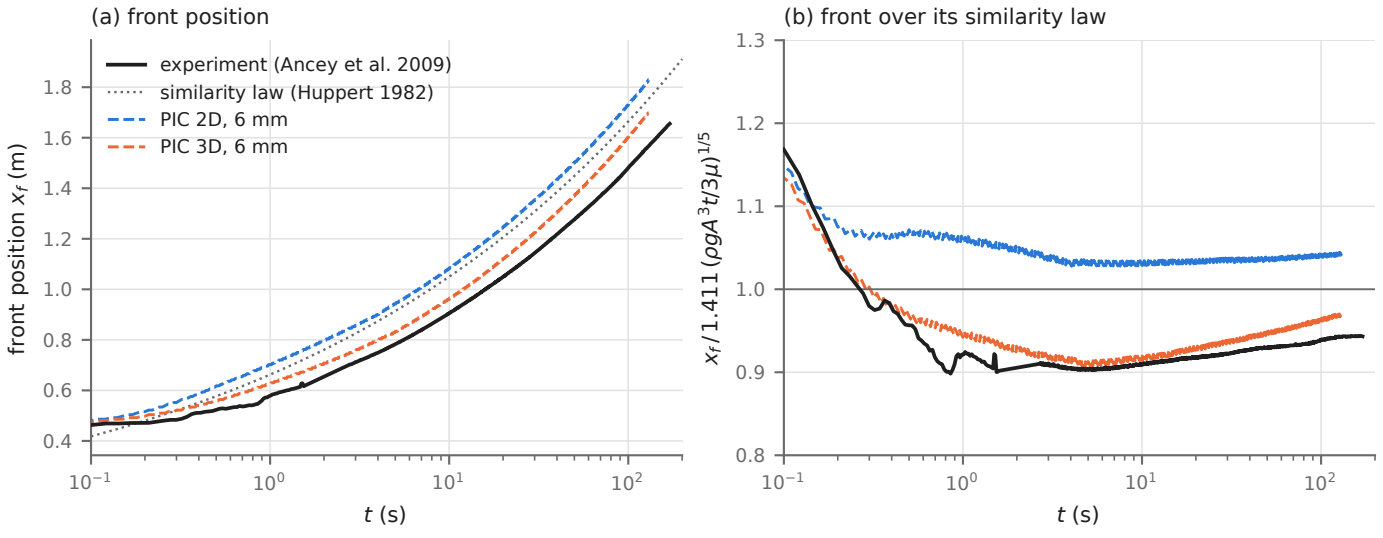
(a) front position
experiment (Ancey et al. 2009)
similarity law (Huppert 1982)
PIC 2D, 6 mm
PIC 3D, 6 mm
front position x_f (m)
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
10^{-1}
10^{0}
10^{1}
10^{2}
t (s)
(b) front over its similarity law
$x_f / 1.411\,(\rho g A^3 t/3\mu)^{1/5}$
1.3
1.2
1.1
1.0
0.9
0.8
10^{-1}
10^{0}
10^{1}
10^{2}
t (s)